

American Option Pricing with Markovian Projections

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① Single-Level L^2 Regression

Formulation

Numerical Results

② Multi Level L^2 Regression

Regression on coarser level

Optimal Transport Map

Black-Scholes Correlated Model

We consider a d -dimensional correlated Black-Scholes model

$$dX(t) = a(t, X(t)) dt + b(t, X(t)) dW(t), \quad 0 \leq t \leq T, \quad X(t) \in \mathbb{R}^d, \quad (1)$$

$$X(0) = x_0. \quad (2)$$

where

- $W(t)$ is a k -dimensional Brownian motion.
- $a(t, x) = r x$, $r = 0.05$.
- $b(t, x) \in \mathbb{R}^{d \times k}$.
- $(b_{\text{BS}}(t, x))_{ij} = x_i \Sigma_{ij}$.

Example ($d, k = 3$)

$$b_{\text{BS}}(t, x) = \text{diag}(x) \text{diag}(\sigma) G, \quad \text{equivalently} \quad (b_{\text{BS}}(t, x))_{ij} = \sigma_i x_i G_{ij}. \quad (3)$$

$$\Sigma_{ij} = \sigma_i G_{ij}, \quad GG^\top = \begin{pmatrix} 1 & 0.8 & 0.3 \\ 0.8 & 1 & 0.1 \\ 0.3 & 0.1 & 1 \end{pmatrix}, \quad \sigma = (0.2, 0.15, 0.1)^\top. \quad (4)$$

$$dX_i(t) = rX_i(t) dt + \sigma_i X_i(t) \sum_{j=1}^k G_{ij} dW_j(t), \quad i = 1, \dots, d. \quad (5)$$

Markovian Projection

Let $S(t) = P_1 X(t)$, with $P_1 \in \mathbb{R}^{1 \times d}$ and $X(t) \in \mathbb{R}^{d \times 1}$. Define the Markovian surrogate $\bar{S}^{(x_0)}(t)$ by

$$\begin{cases} d\bar{S}^{(x_0)}(t) = \bar{a}^{(x_0)}(t, \bar{S}^{(x_0)}(t)) dt + \bar{b}^{(x_0)}(t, \bar{S}^{(x_0)}(t)) d\bar{W}(t), & t \in [0, T], \\ \bar{S}^{(x_0)}(0) = P_1 x_0. \end{cases} \quad (6)$$

The drift and diffusion coefficients are given by conditional expectations:

$$\bar{a}^{(x_0)}(t, s) = \mathbb{E}[P_1 a(t, X(t)) \mid P_1 X(t) = s, X(0) = x_0] \quad (7)$$

$$= r s, \quad (8)$$

and

$$(\bar{b}^{(x_0)}(t, s))^2 = \mathbb{E}\left[P_1 b_{\text{BS}}(t, X(t)) b_{\text{BS}}(t, X(t))^{\top} P_1^{\top} \mid P_1 X(t) = s, X(0) = x_0\right]. \quad (9)$$

For $P_1 = \frac{1}{d}(1, \dots, 1)$ and $C = GG^{\top}$, this becomes

$$(\bar{b}^{x_0}(t, s))^2 = \mathbb{E}\left[\frac{1}{d^2} \sum_{i=1}^d \sum_{j=1}^d \sigma_i \sigma_j X_i(t) X_j(t) C_{ij} \mid P_1 X(t) = s, X(0) = x_0\right]. \quad (10)$$

L^2 Regression Problem

We approximate $(\bar{b}^{x_0})^2(\cdot, \cdot)$ by solving the time-integrated L^2 problem

$$(\bar{b}^{x_0})^2(\cdot, \cdot) \approx \arg \min_{h \in V} \int_0^T \mathbb{E} \left[\left(P b_{\text{BS}}(t, X(t)) b_{\text{BS}}(t, X(t))^\top P^\top - h(t, PX(t)) \right)^2 \middle| X(0) = x_0 \right] dt. \quad (11)$$

For example, one may choose a parametric class

$$V = \{ h : h(t, x) = c_0 + c_1 t + c_2 x \}.$$

Discretizing time with a grid $0 = t_0 < t_1 < \dots < t_N = T$, an empirical Monte Carlo objective is

$$(\bar{b}^{x_0})^2 \approx \arg \min_{h \in V} \frac{1}{M} \sum_{m=1}^M \frac{1}{N} \sum_{n=1}^N \left(\frac{1}{d^2} \sum_{i=1}^d \sum_{j=1}^d \sigma_i \sigma_j X_i^{(m)}(t_n) X_j^{(m)}(t_n) C_{ij} - h(t_n, PX^{(m)}(t_n)) \right)^2. \quad (12)$$

Ansatz:

$$\tilde{b}^2(t, s) \approx \sum_{p=1}^P c_p \psi_p(t, s). \quad (13)$$

Let $\{\tilde{P}_n(x)\}_{n \in \mathbb{N}_0}$ be the orthonormalized Legendre Polynomials on $[-1, 1]$. Define

$$\psi_{(i_1, i_2)}(t, s) = \tilde{P}_{i_1}(\tau(t)) \tilde{P}_{i_2}(\xi(s)), \quad p = (i_1, i_2) \in \Lambda. \quad (14)$$

Consider a **total-degree** index set

$$\Lambda_{d_{\max}} = \{(i_1, i_2) \in \mathbb{N}_0^2 : i_1 + i_2 \leq d_{\max}\}. \quad (15)$$

Using $\{\psi_p\}_{p \in \Lambda}$ the L^2 regression problem becomes

$$\arg \min_{h \in V} \frac{1}{M} \frac{1}{N} \sum_{m=1}^M \sum_{n=1}^N \left(\frac{1}{d^2} \sum_{i=1}^d \sum_{j=1}^d \sigma_i \sigma_j X_i^{(m)}(t_n) X_j^{(m)}(t_n) C_{ij} - \sum_{p \in \Lambda} c_p \psi_p(t_n, PX^{(m)}(t_n)) \right)^2. \quad (16)$$

Design Matrix

$$D \in \mathbb{R}^{MN_t \times |\Lambda|}, D_{m,n,p} = \tilde{P}_{i_1}(\tau(t_n)) \times \tilde{P}_{i_2}(\xi(S_{m,n})) \quad (17)$$

$$\tau(t) = \frac{2t}{T} - 1 \in [-1, 1], \quad \xi(s) = \frac{2(s - s_{\min})}{s_{\max} - s_{\min}} - 1 \in [-1, 1] \quad (18)$$

Target Vector

$$\Psi \in \mathbb{R}^{MN_t}, \Psi_{m,n} = \frac{1}{d^2} \sum_{i=1}^d \sum_{j=1}^d \sigma_i \sigma_j X_i^{(m)}(t_n) X_j^{(m)}(t_n) C_{ij}. \quad (19)$$

Coefficient Vector

$$c = \{c_p\}_{p \in \Lambda} \quad (20)$$

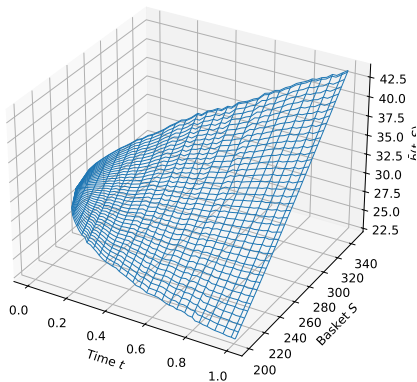
Solves the least-squares regression problem using QR decomposition.

$$D = QR, \quad (21)$$

$$Rc = Q^\top \Psi. \quad (22)$$

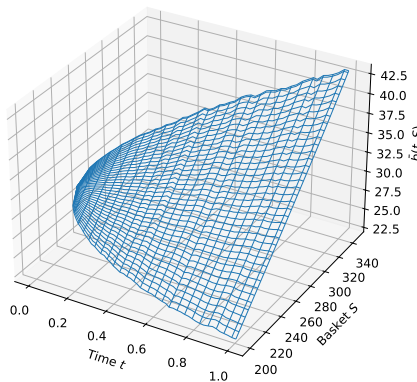
Numerical Results

Single Level Volatility Surface (max degree = 2)



(a) $M = 400, N = 200, dt = 0.005$

Single Level Volatility Surface (max degree = 3)



(b) $M = 400, N = 200, dt = 0.005$

Weak error

$$\varepsilon_{\text{weak}} = \frac{|\mathbb{E}[g(S_T)] - \mathbb{E}[g(\bar{S}_T)]|}{|\mathbb{E}[g(S_T)]|}, \quad (23)$$

where:

- $S_T = P_1^\top X_T$ is the terminal basket value of the true model,
- $\bar{S}_T$ is the terminal value of the projected one-dimensional process,
- g is a payoff function.

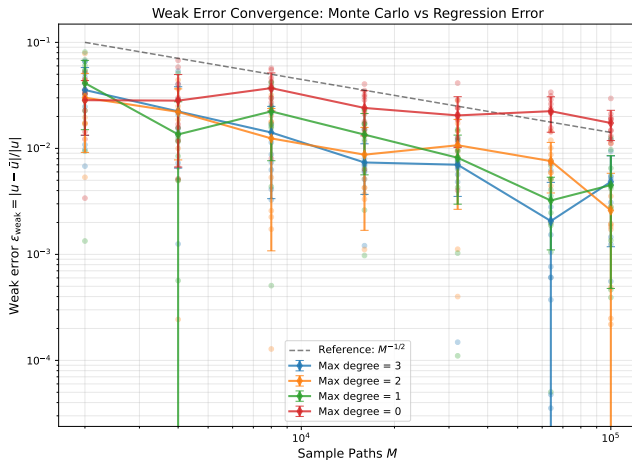
In this implementation, the payoff is the at-the-money European call

$$g(x) = (x - K)^+, \quad K = S_0 = P_1^\top x_0. \quad (24)$$

So the weak error becomes

$$\varepsilon_{\text{weak}} = \frac{|\mathbb{E}[(S_T - K)^+] - \mathbb{E}[(\bar{S}_T - K)^+]|}{|\mathbb{E}[(S_T - K)^+]|}. \quad (25)$$

Numerical Results



Figure

Recall:

$$(\bar{b}^{x_0})^2 \approx \arg \min_{h \in V} \frac{1}{M} \sum_m \frac{1}{N} \sum_n \left(P_1^\top b(t_n, X^m(t_n)) b(t_n, X^m(t_n))^\top P_1 - h(t_n, P_1 X^m(t_n)) \right)^2 \quad (26)$$

where X^m are independent realizations of X . Denote

$$g := P_1^\top b(t, X(t)) b(t, X(t))^\top P_1. \quad (27)$$

$$S_L(f) := \Pi_{V_L} g_0(X_0) + \sum_{\ell=1}^L \Pi_{V_{L-\ell}} (g(X^\ell) - g(X^{\ell-1})) \quad (28)$$

$$\Pi_{V_{L-\ell}} (g(X^\ell) - g(X^{\ell-1})) = \arg \min_{v \in V_{L-\ell}} \left\| (g(X^\ell) - g(X^{\ell-1})) - v(\cdot) \right\|. \quad (29)$$

- $(V_\ell)_{\ell=0}^L$: Subspaces of (total degree) Legendre Polynomials
- $h_\ell = h_0 2^{-\ell}$

- 1 At level ℓ define coarse and fine discretizations $h_\ell = h_0 2^{-\ell}$, $h_{\ell-1} = 2h_\ell$. Simulate the paths X_ℓ and $X_{\ell-1}$ at the same stochastic sample.
- 2 Compute the projection of correction $\Pi_{V_{L-\ell}}(g(X^\ell) - g(X^{\ell-1}))$ by

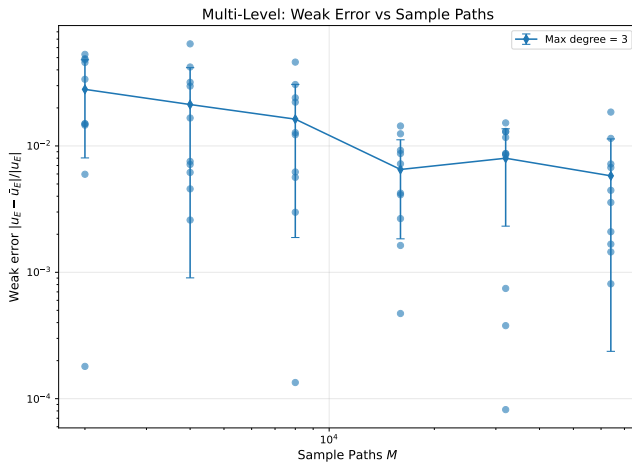
$$\arg \min_{h \in V} \frac{1}{M_\ell} \sum_{m=1}^{M_\ell} \frac{1}{N_{\ell-1}} \sum_{n=1}^{N_{\ell-1}} \left(\left[P^\top b(t_n, X_\ell^m(t_n^{\ell-1})) b(t_n, X_\ell^m(t_n^{\ell-1}))^\top P \right. \right. \\ \left. \left. - P^\top b(t_n, X_{\ell-1}^m(t_n^{\ell-1})) b(t_n, X_{\ell-1}^m(t_n^{\ell-1}))^\top P \right] - h(t_n^{\ell-1}, P X_\ell^m(t_n^{\ell-1})) \right)^2 \quad (30)$$

$$D \in \mathbb{R}^{M_\ell N_{\ell-1} \times |\Lambda|}, \Psi \in \mathbb{R}^{M_\ell N_{\ell-1}}, c_\ell \in \mathbb{R}^{|\Lambda_{L-\ell}|} \quad (31)$$

$$M_\ell = C \times \dim(V_{L-\ell})^2, \quad V_{L-\ell} = \text{span} \left\{ \tilde{P}_i(\tau(t)) \tilde{P}_j(\xi(s)) : i + j \leq L - \ell \right\}. \quad (32)$$

- 3 The estimated coefficient is

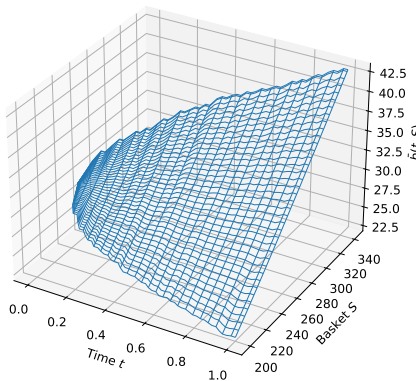
$$(\bar{b}^{x_0})^2(t, s) \approx \sum_{p \in \Lambda_L} c_p \psi(t, s), \quad c = \{c_p\}_{p \in \Lambda_L} = \sum_{\ell=0}^L c_\ell \quad (33)$$



Figure

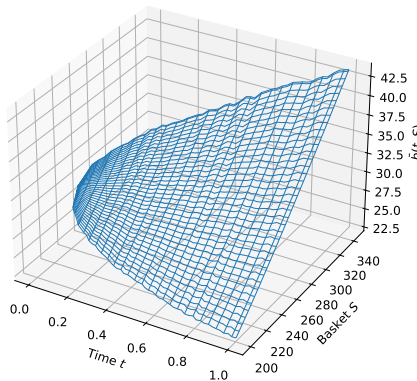
Numerical Results

ML Projected Volatility Surface (max_deg=2)



(a)

ML Projected Volatility Surface (max_deg=3)



(b)

Recall

$$\arg \min_{h \in V} \frac{1}{M_\ell} \sum_{m=1}^{M_\ell} \frac{1}{N_{\ell-1}} \sum_{n=1}^{N_{\ell-1}} \left(\left[P^\top b(t_n, X_\ell^m(t_n^{\ell-1})) b(t_n, X_\ell^m(t_n^{\ell-1}))^\top P \right. \right. \\ \left. \left. - P^\top b(t_n, X_{\ell-1}^m(t_n^{\ell-1})) b(t_n, X_{\ell-1}^m(t_n^{\ell-1}))^\top P \right] - h(t_n^{\ell-1}, P X_\ell^m(t_n^{\ell-1})) \right)^2 \quad (34)$$

For $t_n^{\ell-1}$ consider

$$\mu_{\ell,n} = \mathcal{L}(X_\ell(t_n^{\ell-1})), \quad \mu_{\ell-1,n} = \mathcal{L}(X_{\ell-1}(t_n^{\ell-1})) \quad (35)$$

Assume that there exists a map $T_{\ell,\ell-1,n} : \mathbb{R}^d \rightarrow \mathbb{R}^d$ such that

$$\hat{X}_{\ell-1}^m(t_n^{\ell-1}) = T_{\ell,\ell-1,n}(X_\ell^m(t_n^{\ell-1})) \quad \text{and} \quad (T_{\ell,\ell-1,n})_{\#} \mu_{\ell,n} = \mu_{\ell-1,n} \quad (36)$$

Then the projection of the correction at level ℓ would be on the fine paths X_ℓ , namely

$$g(X_\ell^m(t_n^{\ell-1})) - g(T_{\ell,\ell-1,n}(X_\ell^m(t_n^{\ell-1}))) \quad (37)$$

Optimal Transport Map: Two-Level Example

Data available: Fix a coarse time node t_n^0 .

$$\left\{ X_0^m(t_n^0) \right\}_{m=1}^{M_0}, \quad \left\{ g(X_0^m(t_n^0)) \right\}_{m=1}^{M_0}, \quad (38)$$

and the paired samples

$$\left\{ (X_0^m(t_n^0), X_1^m(t_n^0)) \right\}_{m=1}^{M_1}, \quad \left\{ (g(X_0^m(t_n^0)), g(X_1^m(t_n^0))) \right\}_{m=1}^{M_1}. \quad (39)$$

Goal: approximate $g(X_1(t_n^0))$ using two levels.

Level 0 regression:

$$v_0^{(n)} = \arg \min_{v \in V_0} \mathbb{E} \left[\left(g(X_0(t_n^0)) - v(X_0(t_n^0)) \right)^2 \right]. \quad (40)$$

Transport map from level 1 to level 0:

$$\phi_{10}^{(n)} = \arg \min_{\omega \in W_{10}} \mathbb{E} \left[\left\| X_0(t_n^0) - \omega(X_1(t_n^0)) \right\|^2 \right]. \quad (41)$$

Level 1 correction regression:

$$v_{01}^{(n)} = \arg \min_{v \in V_{01}} \mathbb{E} \left[\left(g(X_1(t_n^0)) - g(\phi_{10}^{(n)}(X_1(t_n^0))) - v(X_1(t_n^0)) \right)^2 \right]. \quad (42)$$

Two-level approximation:

$$g(X_1(t_n^0)) \approx v_0^{(n)}(\phi_{10}^{(n)}(X_1(t_n^0))) + v_{01}^{(n)}(X_1(t_n^0)). \quad (43)$$

Since X_ℓ and $X_{\ell-1}$ are simulated using coupled randomness, it follows that

$$\phi_{10}^{(n)}(x) = x + r^{(n)}(t_n, x) \quad (44)$$

One idea to build the transport map is to consider $S(t) = P \cdot X(t)$. For time $t_n^{\ell-1}$:

① Collect Samples

$$S_f^m = P \cdot X_\ell^m(t_n^{\ell-1}), \quad S_c^m = P \cdot X_{\ell-1}^m(t_n^{\ell-1}) \quad (45)$$

② Build the OT map in 1D using quantiles

- (a) Order $\{S_f^m\} \rightarrow S_{f,i}$ and $\{S_c^m\} \rightarrow S_{c,i}$
- (b) Define $T_n(S_{t,i}) = S_{c,i}$
- (c) Interpolate (linear/monotone) to evaluate $T_n(S)$ for any S
- (d) Define $\hat{S}_c^m = T_n(S_f^m)$

Some Issues: Recall that

$$\arg \min_{h \in V} \frac{1}{M_\ell} \sum_{m=1}^{M_\ell} \frac{1}{N_{\ell-1}} \sum_{n=1}^{N_{\ell-1}} \left(\left[P^\top b(t_n, X_\ell^m(t_n^{\ell-1})) b(t_n, X_\ell^m(t_n^{\ell-1}))^\top P \right. \right. \\ \left. \left. - P^\top b(t_n, X_{\ell-1}^m(t_n^{\ell-1})) b(t_n, X_{\ell-1}^m(t_n^{\ell-1}))^\top P \right] - h(t_n^{\ell-1}, P X_\ell^m(t_n^{\ell-1})) \right)^2 \quad (46)$$

$$D \in \mathbb{R}^{M_\ell N_{\ell-1} \times |\Lambda|}, \Psi \in \mathbb{R}^{M_\ell N_{\ell-1}}, c_\ell \in \mathbb{R}^{|\Lambda_{L-\ell}|} \quad (47)$$

Optimal Transport: Gaussian Brenier Map

We know that at the coarse time nodes $t_n^{\ell-1}$,

$$\log X \left(t_n^{\ell-1} \right) \sim \mathcal{N} (m_n, C_n), \quad (48)$$

where

$$m_n = \log x_0 + \left(r\mathbf{1} - \frac{1}{2} \begin{pmatrix} \sigma_1^2 \\ \vdots \\ \sigma_d^2 \end{pmatrix} \right) t_n^{\ell-1}, \quad C_n = t_n^{\ell-1} \text{diag}(\sigma) \rho \text{diag}(\sigma), \rho = GG^\top \quad (49)$$

Let $\mu_f, \mu_c \in \mathbb{R}^d$ and $C_f, C_c \in \mathbb{R}^{d \times d}$ be the mean vectors and covariance matrices of $\mathcal{N}(\mu_f, C_f)$ and $\mathcal{N}(\mu_c, C_c)$.

The Brenier map transporting $\mathcal{N}(\mu_f, C_f)$ to $\mathcal{N}(\mu_c, C_c)$ for the quadratic cost is

$$T(x) = \mu_c + A(x - \mu_f),$$

where

$$A = C_f^{-1/2} \left(C_f^{1/2} C_c C_f^{1/2} \right)^{1/2} C_f^{-1/2}.$$

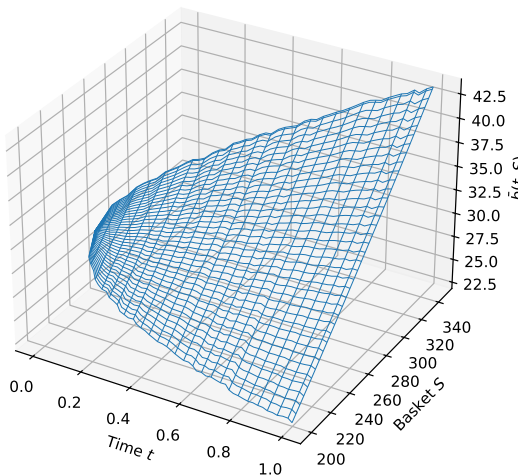
This map satisfies

$$T_{\#} \mathcal{N}(\mu_f, C_f) = \mathcal{N}(\mu_c, C_c).$$

Thus, for each sample m , the transported coarse-state approximation is defined by

$$\hat{X}_{\ell-1}^m(t_n^{\ell-1}) = \exp \left(T_{\ell, \ell-1, n} \left(\log X_{\ell}^m(t_n^{\ell-1}) \right) \right)$$

MLOT Projected Volatility Surface (max_deg=3)



Figure

